

2.1. Recall the mean can be expressed as the functional $T(F) = E_F(X)$ and for the median can be expressed as the functional $T(F) = F^{-1}(0.5)$.

- i) For continuous F , express the trimmed mean and the Winsorized mean in the form of a functional $T(F)$.
- ii) For absolutely continuous F , derive the influence function and gross error sensitivity of the median, and evaluate $E[IF(X; T, F)]$ and $E[IF(X; T, F)^2]$.

2.2.a. Show that the mode is *not* qualitatively robust.

2.2.b. Show that the trimmed mean and the Winsorized mean can be expressed as L-statistics. That is, find the corresponding values of $a_1, \dots, a_n$ in the expression $T_n = \sum_{i=1}^n a_i x_{(i)}$.

2.3. For a location M-estimate, show: (i) $\rho(x, t) = \rho(x - t)$ with $\rho(-r) = \rho(r)$, and (ii) $\psi(x, t) = \psi(x - t)$ with $\psi(-r) = -\psi(r)$.

2.4. Derive the asymptotic relative efficiency of the mean to the median at

- i) the location scale t-family, i.e. when $X \sim \mu + \sigma t_\nu$ where $\mu \in \mathbb{R}, \sigma > 0$ and t_ν is the standard t-distribution on ν degrees of freedom.
- ii) At the mixture of normal distribution $X \sim (1 - \epsilon)Normal(\mu, \sigma^2) + \epsilon Normal(\mu, 9\sigma^2)$.

2.5. Consider a one parameter M-estimator $\hat{\theta}_n$ defined as the minimizer of $\sum_{i=1}^n \rho(x_i; \theta)$. When $\rho(x; \theta)$ corresponds to the negative log-likelihood for a regular parametric family of distributions $F_\theta = F(x; \theta)$, show that $E_F[IF(x; T, F)^2] = 1/I(\theta; X)$ when evaluated at $F = F_\theta$. *Show this directly. Do not use the asymptotic theory of maximum likelihood estimators.*

2.6. Let F denote the distribution function of $X \in \mathbb{R}$. Presume the second moment of F exists, and let σ_F^2 denote the standard deviation of the distribution F .

- i) Express σ_F^2 as a functional, i.e. as $T(F)$.
- ii) Find the influence function of σ_F^2 , i.e. find $IF(x; T, F)$. Also, find its gross error sensitivity and sketch the influence function as a function of $x \in \mathbb{R}$.
- iii) For this case: (a) verify $E_F[IF(X; T, F)] = 0$, and (b) find $E_F[IF(X; T, F)^2]$.
- iv) Let $X_1, \dots, X_n$ be a random sample from F . Presume the fourth moment of F exists, and consider the sample variance $s_n^2 = n^{-1} \sum_{i=1}^n (X_i - \bar{X})^2$. The asymptotic variance of s_n^2 is defined as the variance of the limiting normal distribution of $\sqrt{n}(s_n^2 - \sigma)$. Use the CLT to derive the asymptotic variance of s_n^2 , and verify whether or not it is equal to $E_F[IF(X; T, F)^2]$.

2.7. Let F denote the distribution function of $X \in \mathbb{R}$. Presume the second moment of F exists, and let σ denote the standard deviation of the distribution F .

- i) Express σ as a functional, i.e. as $T(F)$.
- ii) Find the influence function of σ , i.e. find $IF(x; T, F)$. Also, sketch the influence function as a function of $x \in \mathbb{R}$.
- iii) Let $X_1, \dots, X_n$ be a random sample from F . Presume the fourth moment of F exists, and consider the sample standard deviation $s_n = \sqrt{n^{-1} \sum_{i=1}^n (X_i - \bar{X})^2}$. The asymptotic variance of s_n is defined as the variance of the limiting normal distribution of $\sqrt{n}(s_n - \sigma)$. Verify whether or not this asymptotic variance is equal to $E_F[IF(X; T, F)^2]$.